Walmart By Kumud

February 18, 2025

[1]:
```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

# Load dataset
df = pd.read_csv(r"C:\Users\kpswe\Downloads\walmart_data.txt")
df
```

[1]:
```
        User_ID Product_ID Gender    Age  Occupation City_Category  \
0       1000001  P00069042      F   0-17          10             A
1       1000001  P00248942      F   0-17          10             A
2       1000001  P00087842      F   0-17          10             A
3       1000001  P00085442      F   0-17          10             A
4       1000002  P00285442      M    55+          16             C
…         …        …    …    …         …            …
550063  1006033  P00372445      M  51-55          13             B
550064  1006035  P00375436      F  26-35           1             C
550065  1006036  P00375436      F  26-35          15             B
550066  1006038  P00375436      F    55+           1             C
550067  1006039  P00371644      F  46-50           0             B

       Stay_In_Current_City_Years  Marital_Status  Product_Category  Purchase
0                               2               0                 3      8370
1                               2               0                 1     15200
2                               2               0                12      1422
3                               2               0                12      1057
4                              4+               0                 8      7969
…                            …             …                 …        …
550063                          1               1                20       368
550064                          3               0                20       371
550065                         4+               1                20       137
550066                          2               0                20       365
550067                         4+               1                20       490

[550068 rows x 10 columns]
```

[2]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 550068 entries, 0 to 550067
Data columns (total 10 columns):
 #   Column                      Non-Null Count   Dtype
---  ------                      --------------   -----
 0   User_ID                     550068 non-null  int64
 1   Product_ID                  550068 non-null  object
 2   Gender                      550068 non-null  object
 3   Age                         550068 non-null  object
 4   Occupation                  550068 non-null  int64
 5   City_Category               550068 non-null  object
 6   Stay_In_Current_City_Years  550068 non-null  object
 7   Marital_Status              550068 non-null  int64
 8   Product_Category            550068 non-null  int64
 9   Purchase                    550068 non-null  int64
dtypes: int64(5), object(5)
memory usage: 42.0+ MB
```

[3]:
```
df.isna().sum()
```

[3]:
```
User_ID                       0
Product_ID                    0
Gender                        0
Age                           0
Occupation                    0
City_Category                 0
Stay_In_Current_City_Years    0
Marital_Status                0
Product_Category              0
Purchase                      0
dtype: int64
```

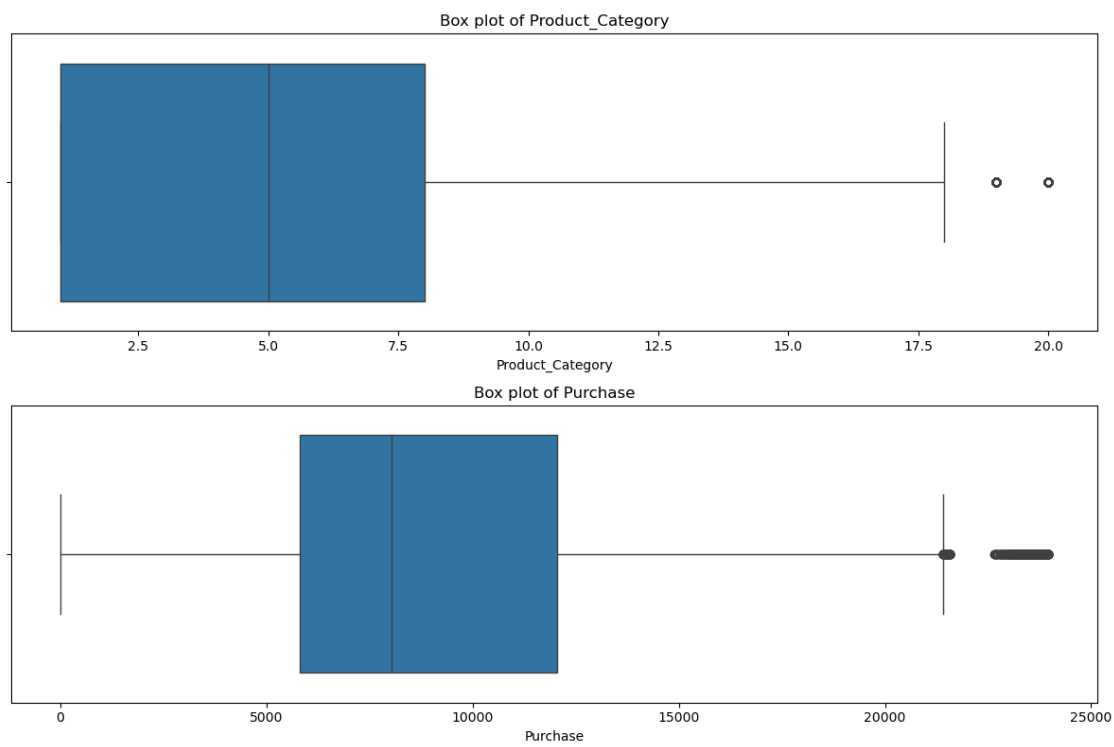
[4]:
```
df.describe()
```

[4]:
```
             User_ID     Occupation  Marital_Status  Product_Category  \
count  5.500680e+05  550068.000000   550068.000000     550068.000000
mean   1.003029e+06       8.076707        0.409653          5.404270
std    1.727592e+03       6.522660        0.491770          3.936211
min    1.000001e+06       0.000000        0.000000          1.000000
25%    1.001516e+06       2.000000        0.000000          1.000000
50%    1.003077e+06       7.000000        0.000000          5.000000
75%    1.004478e+06      14.000000        1.000000          8.000000
max    1.006040e+06      20.000000        1.000000         20.000000

             Purchase
count  550068.000000
```

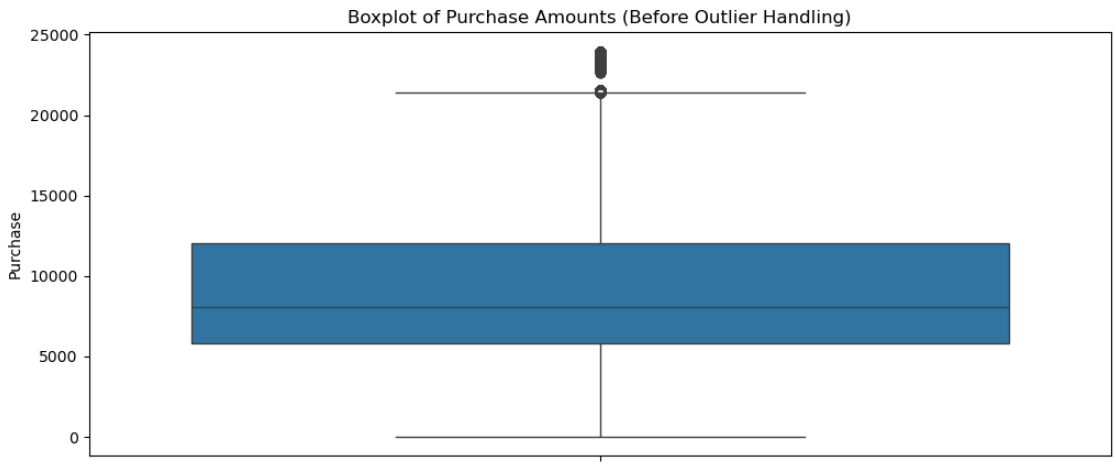
```
mean     9263.968713
std      5023.065394
min        12.000000
25%      5823.000000
50%      8047.000000
75%     12054.000000
max     23961.000000
```

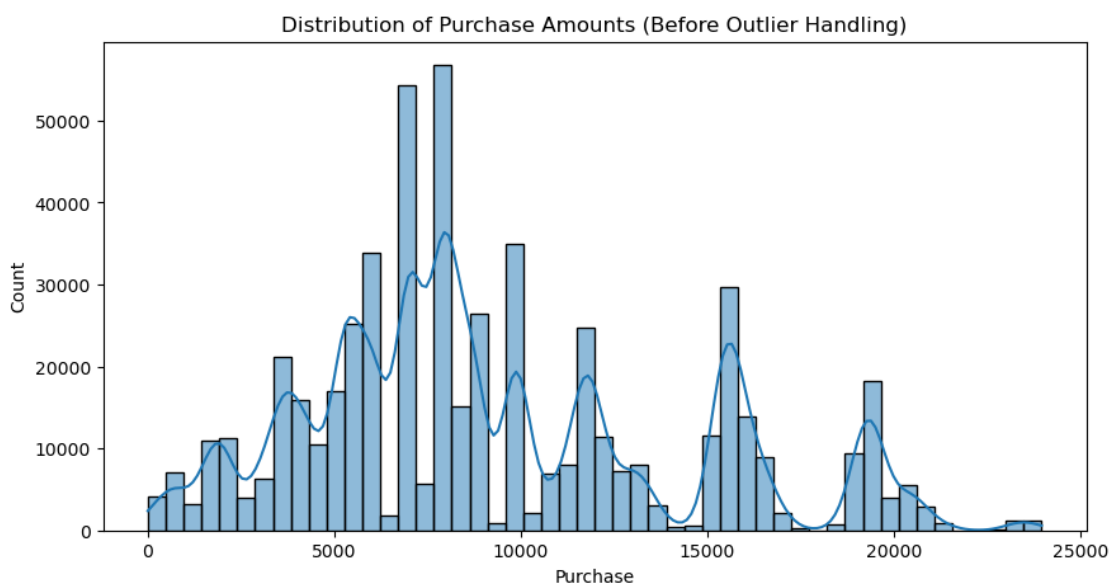
[5]:
```python
df.drop_duplicates(inplace=True)
```

[6]:
```python
numerical_cols = ["Product_Category", "Purchase"]

# Create box plots
plt.figure(figsize=(12, 8))
for i, col in enumerate(numerical_cols, 1):
    plt.subplot(2, 1, i)
    sns.boxplot(x=df[col])
    plt.title(f'Box plot of {col}')

plt.tight_layout()
plt.show()
```

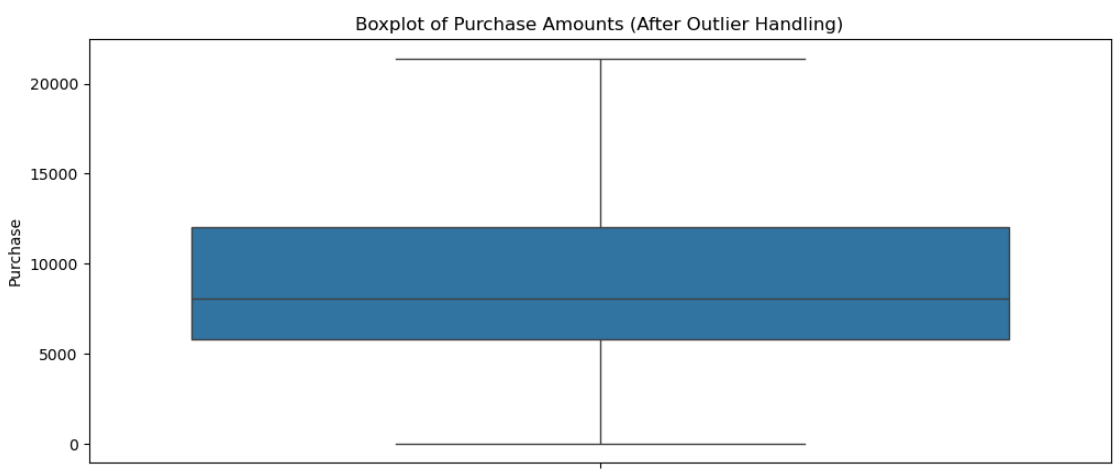


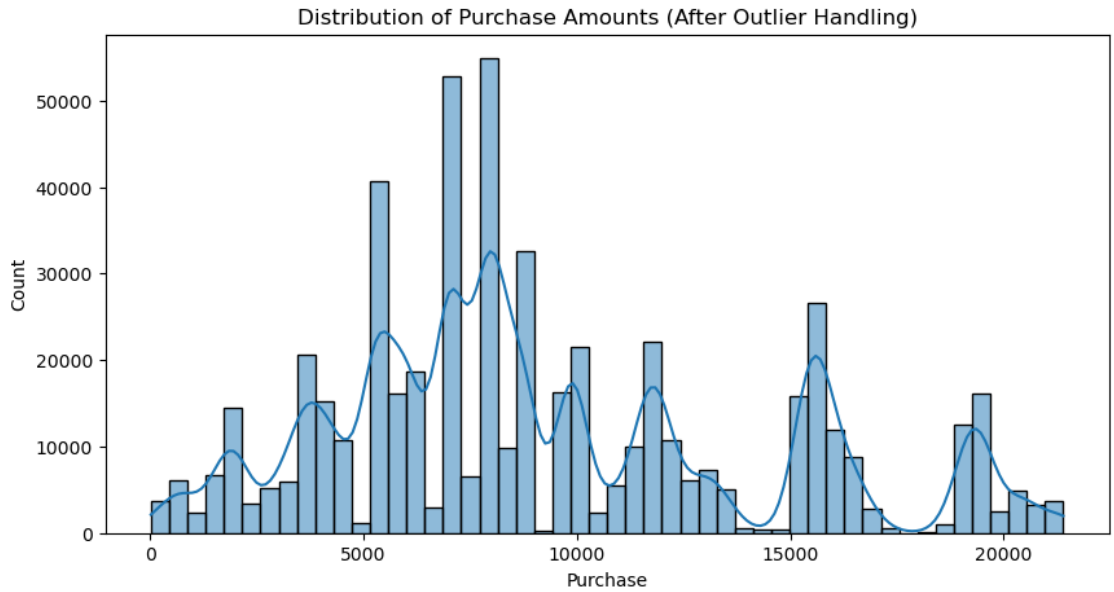
[7]:
```
# Boxplot to detect outliers
plt.figure(figsize=(12, 5))
sns.boxplot(data=df, y="Purchase")
plt.title("Boxplot of Purchase Amounts (Before Outlier Handling)")
plt.show()

# Distribution Plot Before Outlier Handling
plt.figure(figsize=(10, 5))
sns.histplot(df["Purchase"], kde=True, bins=50)
plt.title("Distribution of Purchase Amounts (Before Outlier Handling)")
plt.show()
```





[8]:
```
def handle_outliers_iqr(df, column):
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    df[column] = np.where(df[column] < lower_bound, lower_bound, df[column])
    df[column] = np.where(df[column] > upper_bound, upper_bound, df[column])
    return df

df = handle_outliers_iqr(df, "Purchase")

# Visualizing After Outlier Handling
plt.figure(figsize=(12, 5))
sns.boxplot(data=df, y="Purchase")
plt.title("Boxplot of Purchase Amounts (After Outlier Handling)")
plt.show()

# Distribution Plot After Outlier Handling
plt.figure(figsize=(10, 5))
sns.histplot(df["Purchase"], kde=True, bins=50)
plt.title("Distribution of Purchase Amounts (After Outlier Handling)")
plt.show()
```



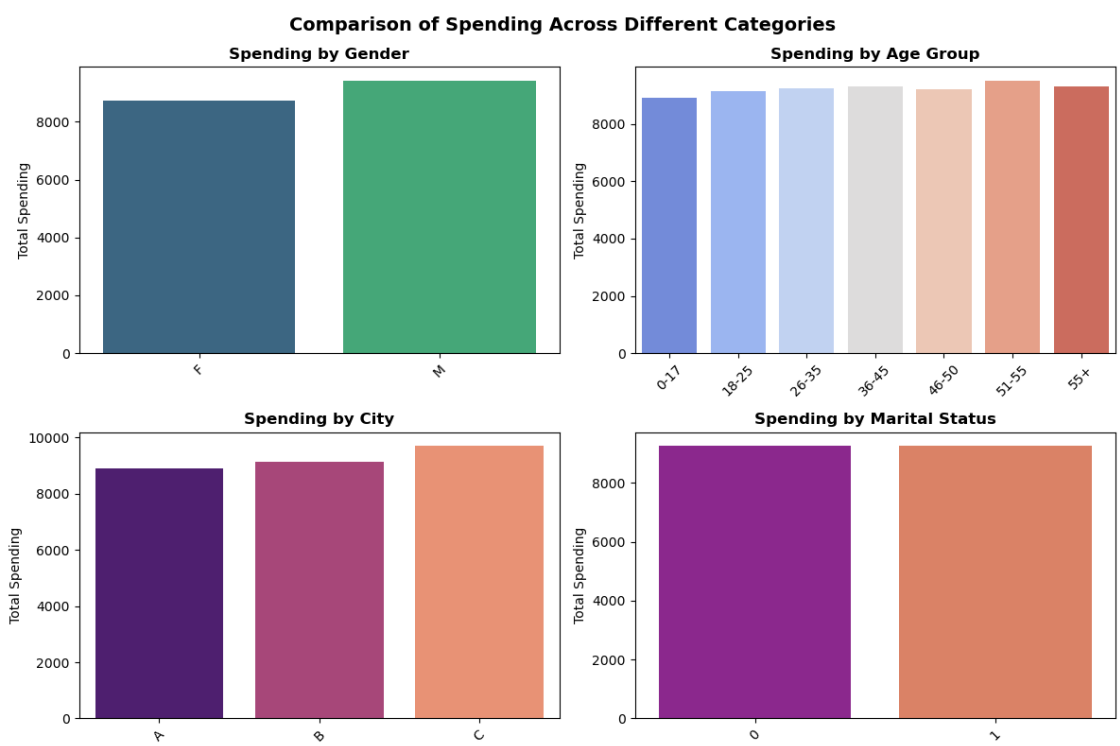


[9]:
```
# Spending by Gender
gender_spending = df.groupby("Gender")["Purchase"].mean()
print("Spending by Gender:\n", gender_spending)
print()
# Spending by Age Group
age_spending = df.groupby("Age")["Purchase"].mean()
print("Spending by Age Group:\n", age_spending)
print()
# Spending by Marital Status
marital_spending = df.groupby("Marital_Status")["Purchase"].mean()
print("Spending by Marital Status:\n", marital_spending)
print()
# Spending by City
city_spending = df.groupby("City_Category")["Purchase"].mean()
print("Spending by City:\n", city_spending)
print()
# Spending by Product Category
product_spending = df.groupby("Product_Category")["Purchase"].mean()
print("Spending by Product:\n", product_spending)
```

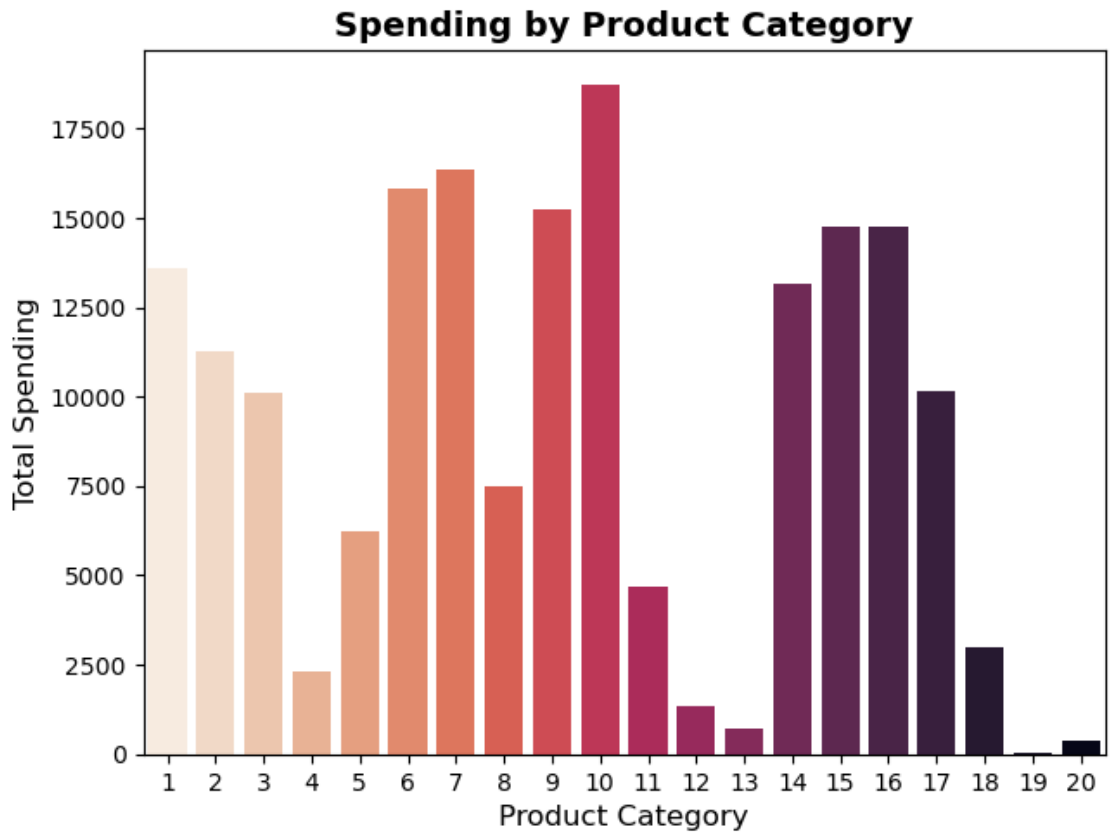
```
Spending by Gender:
 Gender
F    8726.256327
M    9428.373455
Name: Purchase, dtype: float64

Spending by Age Group:
 Age
```

```
0-17     8925.539597
18-25    9164.189554
26-35    9244.947060
36-45    9320.888550
46-50    9198.531093
51-55    9519.560427
55+      9319.768741
Name: Purchase, dtype: float64

Spending by Marital Status:
 Marital_Status
0    9257.517959
1    9251.430702
Name: Purchase, dtype: float64

Spending by City:
 City_Category
A    8903.502451
B    9142.912942
C    9709.786528
Name: Purchase, dtype: float64

Spending by Product:
 Product_Category
1     13606.218596
2     11251.935384
3     10096.705734
4      2329.659491
5      6240.088178
6     15838.478550
7     16365.689600
8      7498.958078
9     15233.089024
10    18745.626829
11     4685.268456
12     1350.859894
13      722.400613
14    13141.625739
15    14775.791017
16    14766.037037
17    10170.759516
18     2972.864320
19       37.041797
20      370.481176
Name: Purchase, dtype: float64
```

[10]:
```
# Define subplots
fig, axes = plt.subplots(2, 2, figsize=(12, 8))

# Data and titles for each plot
data_sets = [gender_spending, age_spending, city_spending, marital_spending]
titles = ["Spending by Gender", "Spending by Age Group", "Spending by City",␣
↪"Spending by Marital Status"]
palettes = ["viridis", "coolwarm", "magma", "plasma"]  # Distinct but visually␣
↪appealing color schemes

# Loop through the axes and plot each dataset
for ax, data, title, palette in zip(axes.flat, data_sets, titles, palettes):
    sns.barplot(x=data.index, y=data.values, hue=data.index, palette=palette,␣
↪legend=False, ax=ax)
    ax.set_title(title, fontsize=12, fontweight="bold")
    ax.set_xlabel("")  # Remove redundant x-labels
    ax.set_ylabel("Total Spending")
    ax.tick_params(axis="x", rotation=45)  # Rotate x-axis labels for better␣
↪readability

# Adjust layout and add a global title
fig.suptitle("Comparison of Spending Across Different Categories", fontsize=14,␣
↪fontweight="bold")
plt.tight_layout()
plt.show()
```



[11]:
```
sns.barplot(x=product_spending.index, y=product_spending.values,␣
↪hue=product_spending.index, palette="rocket_r", legend=False)
plt.tight_layout()
plt.title("Spending by Product Category", fontsize=14, fontweight="bold")
plt.xlabel("Product Category", fontsize=12)
plt.ylabel("Total Spending", fontsize=12)
plt.show()
```



[12]:
```
def confidence_interval(data, confidence=0.95, sample_size=1000):
    sample = np.random.choice(data, size=sample_size, replace=True)
    sample_mean = np.mean(sample)
    sample_std = np.std(sample, ddof=1)  # Sample standard deviation
    z_score = stats.norm.ppf((1 + confidence) / 2)  # Critical value for given␣
↪confidence level

    margin_error = z_score * (sample_std / np.sqrt(sample_size))
    lower_bound = sample_mean - margin_error
```

```
    upper_bound = sample_mean + margin_error
    ci_width = upper_bound - lower_bound

    return lower_bound, upper_bound, sample_mean, ci_width
```

[13]:
```
def confidence_interval(data, confidence=0.95, sample_size=1000):
    sample = np.random.choice(data, size=sample_size, replace=True)
    sample_mean = np.mean(sample)
    sample_std = np.std(sample, ddof=1)  # Sample standard deviation
    z_score = stats.norm.ppf((1 + confidence) / 2)  # Critical value for given␣
 ↪confidence level

    margin_error = z_score * (sample_std / np.sqrt(sample_size))
    lower_bound = sample_mean - margin_error
    upper_bound = sample_mean + margin_error
    ci_width = upper_bound - lower_bound

    return lower_bound, upper_bound, sample_mean, ci_width
```

[19]:
```
# Different sample sizes and confidence levels
sample_sizes = [100, 500, 1000, 5000, 10000]
confidence_levels = [0.90, 0.95, 0.99]

gender_ci_results = []
for sample_size in sample_sizes:
    for conf_level in confidence_levels:
        lower, upper, mean, width = confidence_interval(gender_spending,␣
 ↪confidence=conf_level, sample_size=sample_size)
        gender_ci_results.append([sample_size, conf_level, lower, upper, mean,␣
 ↪width])

gender_ci_df = pd.DataFrame(gender_ci_results, columns=["Sample Size",␣
 ↪"Confidence Level", "Lower", "Upper", "Mean", "Width"])
print("Confidance Interval by Gender:")
print(gender_ci_df)
print()

age_ci_results = []
for sample_size in sample_sizes:
    for conf_level in confidence_levels:
        lower, upper, mean, width = confidence_interval(age_spending,␣
 ↪confidence=conf_level, sample_size=sample_size)
        age_ci_results.append([sample_size, conf_level, lower, upper, mean,␣
 ↪width])

age_ci_df = pd.DataFrame(age_ci_results, columns=["Sample Size", "Confidence␣
 ↪Level", "Lower", "Upper", "Mean", "Width"])
```

```
print("Confidance Interval by Age Group:")
print(age_ci_df)
print()

city_ci_results = []
for sample_size in sample_sizes:
    for conf_level in confidence_levels:
        lower, upper, mean, width = confidence_interval(city_spending,␣
 ↪confidence=conf_level, sample_size=sample_size)
        city_ci_results.append([sample_size, conf_level, lower, upper, mean,␣
 ↪width])

city_ci_df = pd.DataFrame(city_ci_results, columns=["Sample Size", "Confidence␣
 ↪Level", "Lower", "Upper", "Mean", "Width"])
print("Confidance Interval by City:")
print(city_ci_df)
print()

marital_ci_results = []
for sample_size in sample_sizes:
    for conf_level in confidence_levels:
        lower, upper, mean, width = confidence_interval(marital_spending,␣
 ↪confidence=conf_level, sample_size=sample_size)
        marital_ci_results.append([sample_size, conf_level, lower, upper, mean,␣
 ↪width])

marital_ci_df = pd.DataFrame(marital_ci_results, columns=["Sample Size",␣
 ↪"Confidence Level", "Lower", "Upper", "Mean", "Width"])
print("Confidance Interval by Marial Status:")
print(marital_ci_df)
print()

product_ci_results = []
for sample_size in sample_sizes:
    for conf_level in confidence_levels:
        lower, upper, mean, width = confidence_interval(product_spending,␣
 ↪confidence=conf_level, sample_size=sample_size)
        product_ci_results.append([sample_size, conf_level, lower, upper, mean,␣
 ↪width])

product_ci_df = pd.DataFrame(product_ci_results, columns=["Sample Size",␣
 ↪"Confidence Level", "Lower", "Upper", "Mean", "Width"])
print("Confidance Interval by Product:")
print(product_ci_df)
```

```
Confidance Interval by Gender:
    Sample Size  Confidence Level        Lower        Upper         Mean  \
```

0	100	0.90	9068.999747	9183.926432	9126.463090
1	100	0.95	8966.534722	9103.841004	9035.187863
2	100	0.99	9014.808619	9195.990533	9105.399576
3	500	0.90	9065.528159	9117.186308	9091.357233
4	500	0.95	9029.697692	9091.230467	9060.464080
5	500	0.99	9034.027170	9114.985675	9074.506422
6	1000	0.90	9038.714858	9075.192131	9056.953494
7	1000	0.95	9058.354733	9101.891985	9080.123359
8	1000	0.99	9035.385589	9092.563741	9063.974665
9	5000	0.90	9075.327736	9091.659307	9083.493522
10	5000	0.95	9072.218070	9091.679657	9081.948864
11	5000	0.99	9066.631928	9092.210557	9079.421242
12	10000	0.90	9072.663620	9084.212937	9078.438278
13	10000	0.95	9067.836280	9081.597835	9074.717057
14	10000	0.99	9074.030346	9092.114157	9083.072251

	Width
0	114.926685
1	137.306282
2	181.181913
3	51.658149
4	61.532775
5	80.958504
6	36.477273
7	43.537252
8	57.178152
9	16.331572
10	19.461587
11	25.578629
12	11.549317
13	13.761554
14	18.083811

Confidance Interval by Age Group:

	Sample Size	Confidence Level	Lower	Upper	Mean \
0	100	0.90	9220.877898	9274.974345	9247.926121
1	100	0.95	9194.379333	9253.807610	9224.093472
2	100	0.99	9160.623393	9261.328007	9210.975700
3	500	0.90	9236.134335	9260.806418	9248.470377
4	500	0.95	9226.927887	9256.522750	9241.725318
5	500	0.99	9213.424086	9253.325775	9233.374930
6	1000	0.90	9235.432355	9252.889094	9244.160725
7	1000	0.95	9223.003061	9244.245178	9233.624120
8	1000	0.99	9224.059271	9250.766429	9237.412850
9	5000	0.90	9235.083402	9242.903981	9238.993692
10	5000	0.95	9238.009538	9247.308080	9242.658809
11	5000	0.99	9236.207953	9248.469033	9242.338493
12	10000	0.90	9234.797313	9240.379385	9237.588349

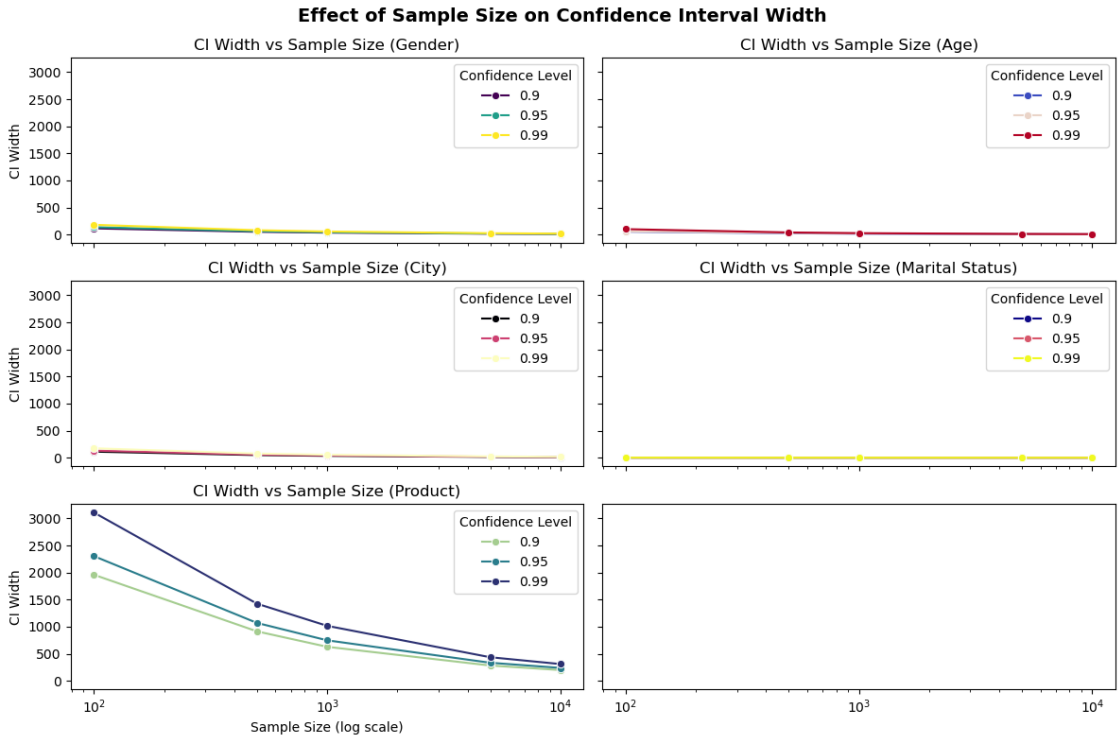
```
13        10000              0.95  9237.389310  9243.983380  9240.686345
14        10000              0.99  9237.883077  9246.712361  9242.297719

         Width
0    54.096447
1    59.428277
2   100.704614
3    24.672083
4    29.594862
5    39.901689
6    17.456739
7    21.242117
8    26.707158
9     7.820579
10    9.298542
11   12.261081
12    5.582072
13    6.594070
14    8.829284

Confidance Interval by City:
    Sample Size  Confidence Level        Lower        Upper         Mean  \
0           100              0.90  9208.408594  9321.006086  9264.707340
1           100              0.95  9169.154938  9300.545225  9234.850082
2           100              0.99  9190.082383  9365.034398  9277.558391
3           500              0.90  9231.392185  9280.840182  9256.116184
4           500              0.95  9225.439152  9283.821300  9254.630226
5           500              0.99  9193.825686  9271.086268  9232.455977
6          1000              0.90  9229.328634  9264.433441  9246.881038
7          1000              0.95  9246.141639  9288.639061  9267.390350
8          1000              0.99  9231.002745  9286.116821  9258.559783
9          5000              0.90  9235.972063  9251.552215  9243.762139
10         5000              0.95  9251.715661  9270.513479  9261.114570
11         5000              0.99  9242.198268  9266.697775  9254.448022
12        10000              0.90  9242.364232  9253.449062  9247.906647
13        10000              0.95  9244.315223  9257.547677  9250.931450
14        10000              0.99  9239.759349  9257.144759  9248.452054

         Width
0   112.597492
1   131.390287
2   174.952015
3    49.447997
4    58.382149
5    77.260582
6    35.104807
7    42.497422
8    55.114076
```

```
9    15.580153
10   18.797818
11   24.499507
12   11.084830
13   13.232454
14   17.385411

Confidance Interval by Marial Status:
    Sample Size  Confidence Level        Lower        Upper         Mean  \
0           100              0.90  9253.910404  9254.916512  9254.413458
1           100              0.95  9253.454582  9254.641863  9254.048222
2           100              0.99  9252.978952  9254.508767  9253.743860
3           500              0.90  9254.068003  9254.515423  9254.291713
4           500              0.95  9254.025146  9254.558280  9254.291713
5           500              0.99  9253.977680  9254.678793  9254.328236
6          1000              0.90  9254.370748  9254.687483  9254.529116
7          1000              0.95  9254.291681  9254.669154  9254.480418
8          1000              0.99  9254.329915  9254.825713  9254.577814
9          5000              0.90  9254.392567  9254.534180  9254.463373
10         5000              0.95  9254.361961  9254.530698  9254.446329
11         5000              0.99  9254.407287  9254.629031  9254.518159
12        10000              0.90  9254.424265  9254.524396  9254.474331
13        10000              0.95  9254.385457  9254.504766  9254.445112
14        10000              0.99  9254.451335  9254.608114  9254.529725

       Width
0   1.006108
1   1.187282
2   1.529815
3   0.447420
4   0.533134
5   0.701112
6   0.316735
7   0.377473
8   0.495798
9   0.141613
10  0.168737
11  0.221744
12  0.100131
13  0.119309
14  0.156779

Confidance Interval by Product:
    Sample Size  Confidence Level        Lower         Upper         Mean  \
0           100              0.90  7211.972419   9170.417109  8191.194764
1           100              0.95  7900.931136  10203.196257  9052.063697
2           100              0.99  7393.358787  10500.607579  8946.983183
3           500              0.90  8187.486433   9101.426181  8644.456307
```

```
4           500              0.95  8635.781959   9705.215903  9170.498931
5           500              0.99  8120.628098   9545.430082  8833.029090
6          1000              0.90  8812.657875   9441.468887  9127.063381
7          1000              0.95  8531.216439   9279.663879  8905.440159
8          1000              0.99  8138.777771   9154.702518  8646.740145
9          5000              0.90  8795.934509   9079.516599  8937.725554
10         5000              0.95  8906.175997   9240.738345  9073.457171
11         5000              0.99  8936.164654   9372.719247  9154.441950
12        10000              0.90  8969.624629   9169.475951  9069.550290
13        10000              0.95  8839.649690   9078.783179  8959.216435
14        10000              0.99  8723.284239   9033.774146  8878.529193

          Width
0   1958.444690
1   2302.265121
2   3107.248791
3    913.939748
4   1069.433944
5   1424.801984
6    628.811012
7    748.447440
8   1015.924747
9    283.582091
10   334.562348
11   436.554593
12   199.851322
13   239.133489
14   310.489907
```

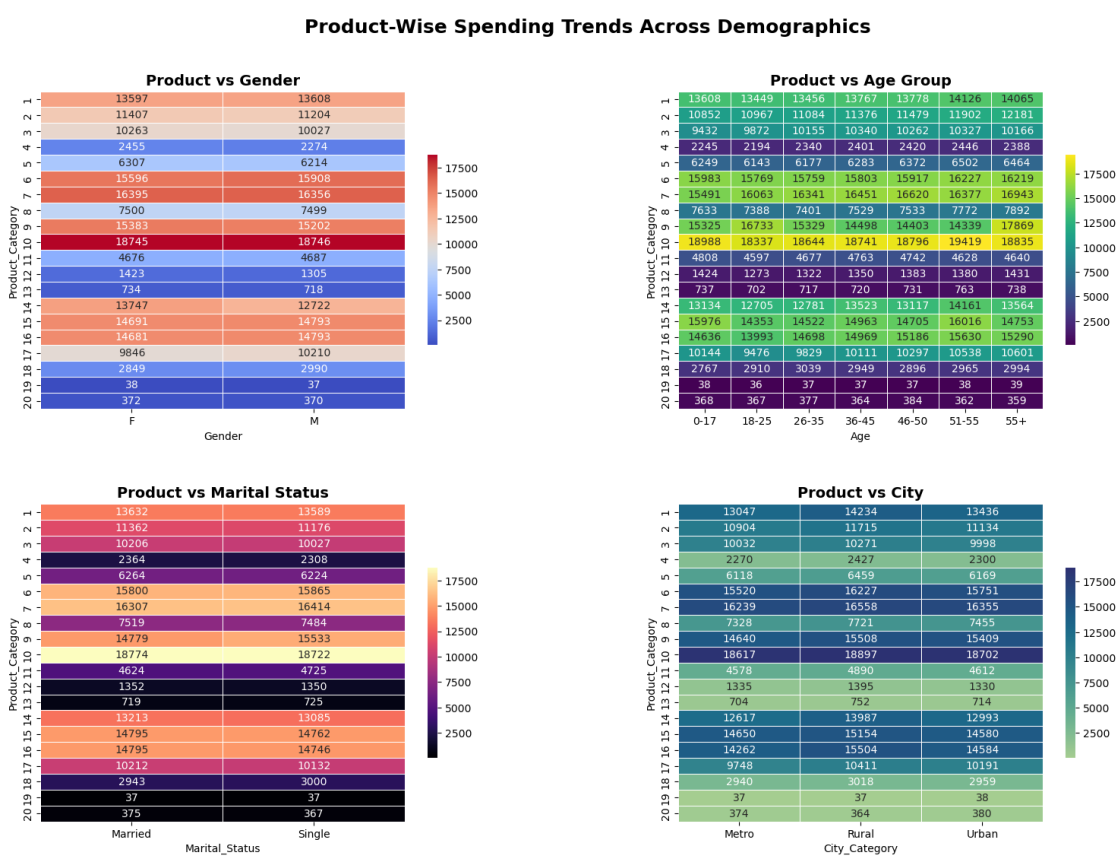
[20]:
```
fig, axes = plt.subplots(3, 2, figsize=(12, 8), sharex=True, sharey=True)

# CI width vs. Sample size plots for different groupings
ci_dfs = [gender_ci_df, age_ci_df, city_ci_df, marital_ci_df,product_ci_df]
titles = ["Gender", "Age", "City", "Marital Status","Product"]
palettes = ["viridis", "coolwarm", "magma", "plasma","crest"]  # Keeping␣
 ↪diverse but distinct palettes

for ax, ci_df, title, palette in zip(axes.flat, ci_dfs, titles, palettes):
    sns.lineplot(data=ci_df, x="Sample Size", y="Width", hue="Confidence␣
 ↪Level", marker="o", palette=palette, ax=ax)
    ax.set_xscale("log")  # Log scale for better visualization
    ax.set_title(f"CI Width vs Sample Size ({title})")  # Subplot-specific title
    ax.set_xlabel("Sample Size (log scale)")
    ax.set_ylabel("CI Width")
    ax.legend(title="Confidence Level", loc="upper right")  # Per subplot legend

# Overall figure title
```

```
fig.suptitle("Effect of Sample Size on Confidence Interval Width", fontsize=14,␣
↪fontweight="bold")

plt.tight_layout()
plt.show()
```



[17]:
```
# Replace category labels for better readability
df["Marital_Status"] = df["Marital_Status"].replace({0: "Single", 1: "Married"})
df["City_Category"] = df["City_Category"].replace({"A": "Metro", "B": "Urban",␣
↪"C": "Rural"})

# Compute average purchase per category
grouped_data = {
    "Gender": df.groupby(["Product_Category", "Gender"])["Purchase"].mean().
↪unstack(),
    "Age Group": df.groupby(["Product_Category", "Age"])["Purchase"].mean().
↪unstack(),
    "Marital Status": df.groupby(["Product_Category",␣
↪"Marital_Status"])["Purchase"].mean().unstack(),
    "City": df.groupby(["Product_Category", "City_Category"])["Purchase"].
↪mean().unstack()
}
```

```python
# Initialize figure
fig, axes = plt.subplots(2, 2, figsize=(16, 12))

# Define color palettes
palettes = ["coolwarm", "viridis", "magma", "crest"]

# Create heatmaps with improved axis formatting
for i, ((title, data), cmap) in enumerate(zip(grouped_data.items(), palettes)):
    ax = axes[i // 2, i % 2]  # Get subplot position
    heatmap = sns.heatmap(
        data, cmap=cmap, annot=True, fmt=".0f", linewidths=0.5, ax=ax,␣
↪cbar=True,
        cbar_kws={"shrink": 0.6}
    )

    ax.set_title(f"Product vs {title}", fontsize=14, fontweight="bold")

plt.subplots_adjust(left=0.05, right=0.95, top=0.9, bottom=0.1, hspace=0.3,␣
↪wspace=0.4)
plt.suptitle("Product-Wise Spending Trends Across Demographics", fontsize=18,␣
↪fontweight="bold")

plt.show()
```



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